Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Multi Agents System with AutoGen (C1034)

Course Code: C1034

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	11 August 2026	Initial release aligned across slide deck, Learner Guide, Lesson Plan, and eight connected labs.	Tertiary Infotech Trainer

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Course Information

Course Title	Multi Agents System with AutoGen (C1034)
Course Reference	C1034
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 scheduled hours per day (16 hours)
Daily Timing	9:30 am - 6:30 pm (1-hour lunch; two 15-minute tea breaks)
Mode	Instructor-led, concept-first learning with connected hands-on labs
Trainer	Tertiary Infotech Trainer

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain LLM agents, the AgentChat execution model, and the practical differences between AutoGen and CrewAI abstractions.
- LO2: Configure and run an AutoGen AssistantAgent with a model client, clear messages, a typed read-only tool, and inspectable event evidence.
- LO3: Design a Retrieval-Augmented Generation pipeline with source-labelled chunks, transparent ranking, metadata, context limits, and stable retrieval questions.
- LO4: Build and verify a grounded RAG agent that cites retrieved evidence, treats documents as untrusted data, and handles insufficient evidence honestly.
- LO5: Select and implement appropriate multi-agent patterns with specialist roles, explicit handoffs, shared context, and bounded speaker routing.
- LO6: Apply termination, state management, guardrails, observability, regression checks, ownership, and recovery controls to an integrated AutoGen workflow.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

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Course Schedule

Day 1 — AgentChat Foundations and Grounded Retrieval

Time	Duration	Topic / Activity
9:30–9:50	20 min	Welcome, environment check, course roadmap and learning outcomes
9:50–10:50	60 min	**Topic 1 — Introduction to AutoGen and CrewAI (concepts, architecture and worked example)**
10:50–11:05	15 min	Tea break
11:05–12:05	60 min	Hands-on: Lab 1: Verify AutoGen and Run a First Agent
12:05–13:00	55 min	Hands-on: Lab 2: Add a Typed Ticket Lookup Tool
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	**Topic 2 — Retrieval-Augmented Generation (RAG) (concepts, pipeline and worked example)**
15:00–15:15	15 min	Tea break
15:15–16:05	50 min	Hands-on: Lab 3: Prepare a Source-Traceable Knowledge Index
16:05–16:55	50 min	Hands-on: Lab 4: Evaluate Retrieval with Stable Questions
16:55–18:10	75 min	Hands-on: Lab 5: Build a Grounded RAG Agent
18:10–18:30	20 min	Day 1 recap, retrieval clinic and reflection

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

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Day 2 — Multi-Agent Orchestration and Operational Readiness

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap, checkpoint recovery and Day 2 orientation
9:45–10:45	60 min	**Topic 3 — Implementing a Multi-Agent AI Workflow (roles, patterns and handoff contracts)**
10:45–11:00	15 min	Tea break
11:00–12:00	60 min	Hands-on: Lab 6: Design Roles and Run a Round-Robin Team
12:00–13:00	60 min	Hands-on: Lab 7: Route Specialists with SelectorGroupChat
13:00–14:00	60 min	Lunch break
14:00–15:00	60 min	**Topic 3 — Implementing a Multi-Agent AI Workflow (termination, state, observability and operations)**
15:00–15:15	15 min	Tea break
15:15–18:05	170 min	Hands-on: Lab 8: Integrate and Verify the OpsDesk Workflow
18:05–18:30	25 min	Course recap, operational checklist, action plan and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Introduction to AutoGen and CrewAI	2 of 8 labs (25%)	Lab 1, Lab 2
Topic 02: Retrieval-Augmented Generation (RAG)	3 of 8 labs (38%)	Lab 3, Lab 4, Lab 5
Topic 03: Implementing a Multi-Agent AI Workflow	3 of 8 labs (38%)	Lab 6, Lab 7, Lab 8

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